PRÁCTICA 8

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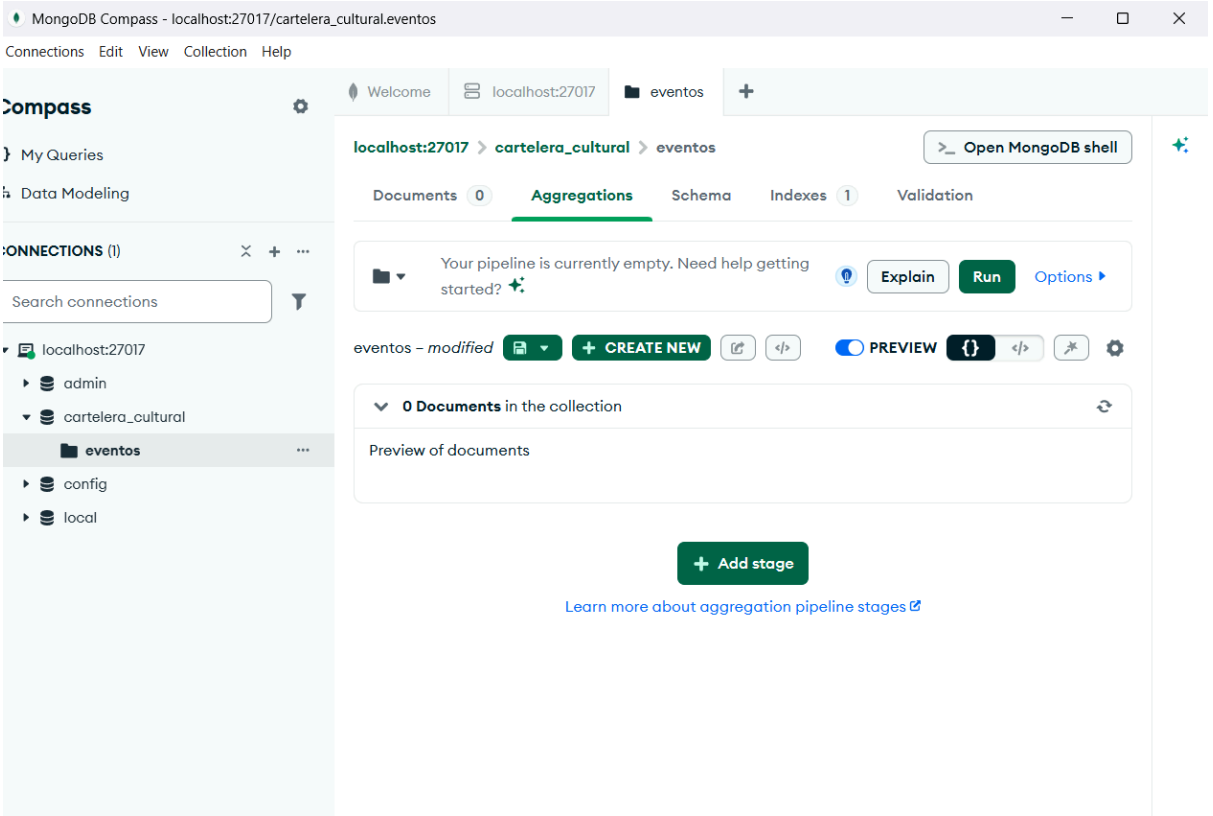
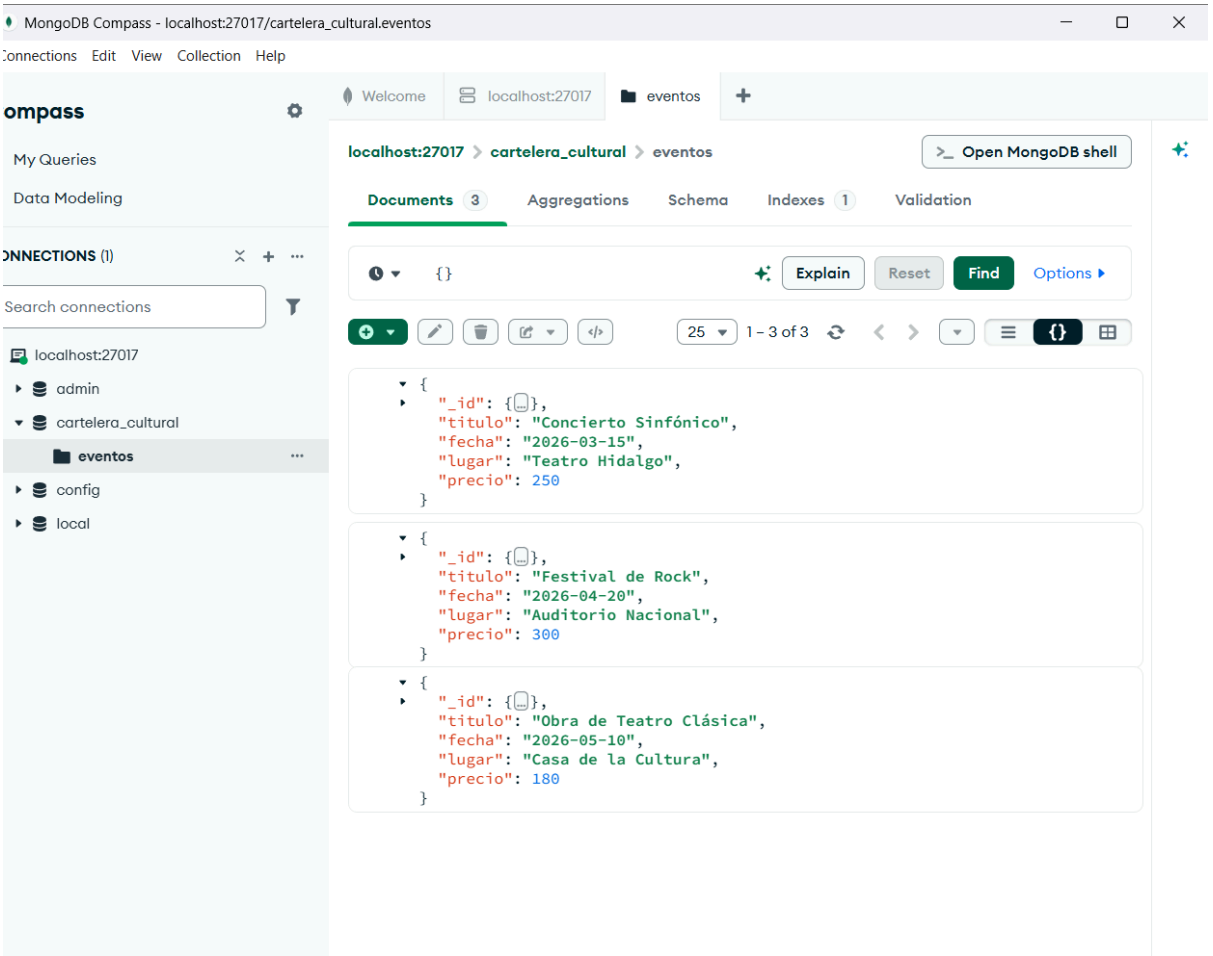
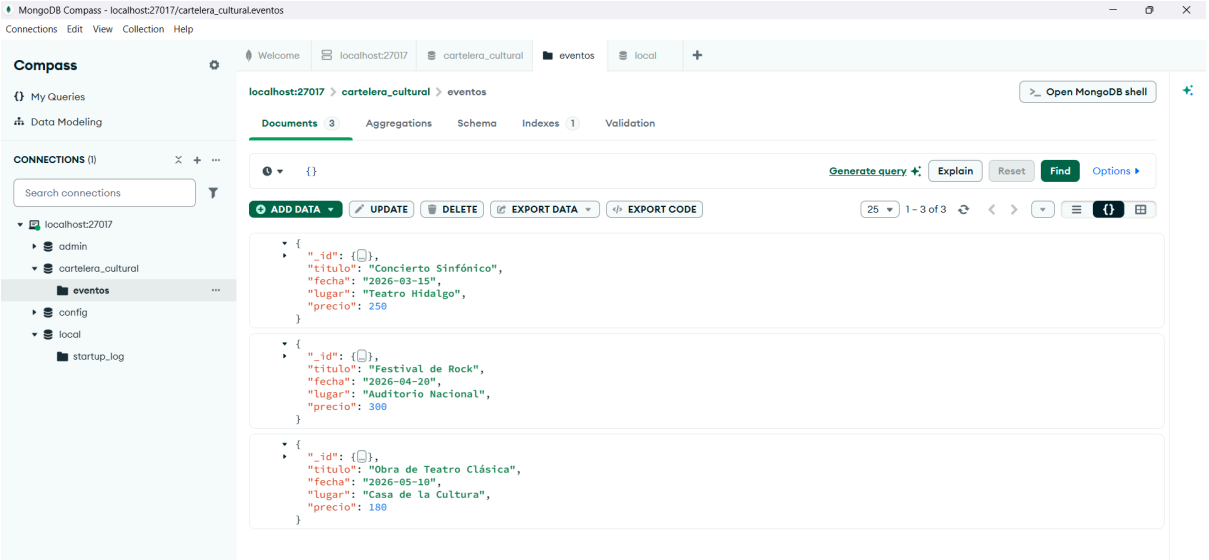
TIID 04_01
Tecnologías de la información e innovación Digital

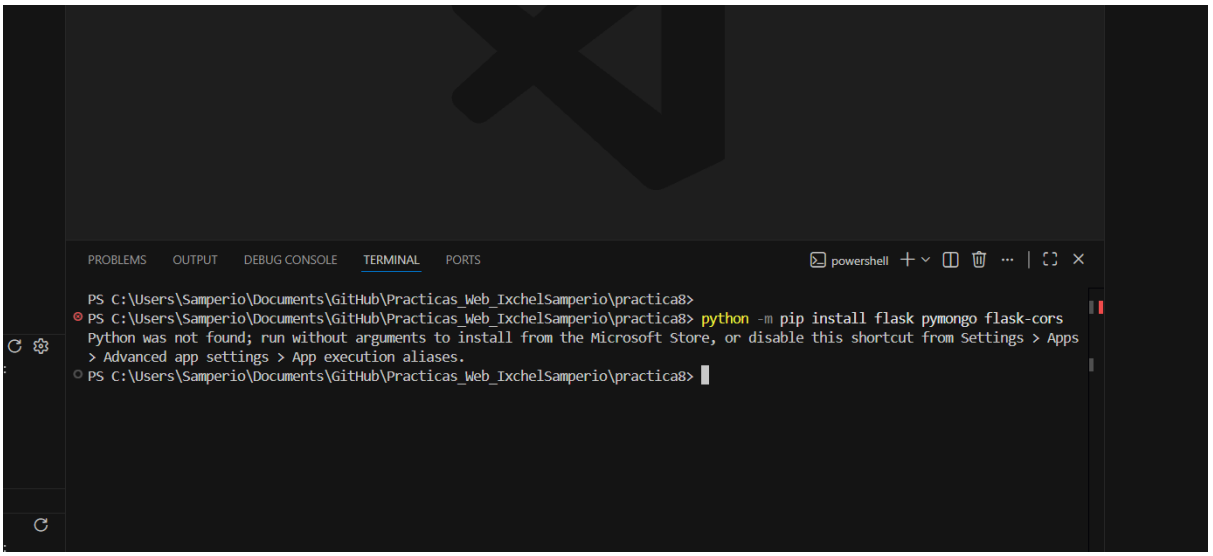
APLICACIONES WEB
Nombre del profesor: Diana Xochitl Ruano Vargas

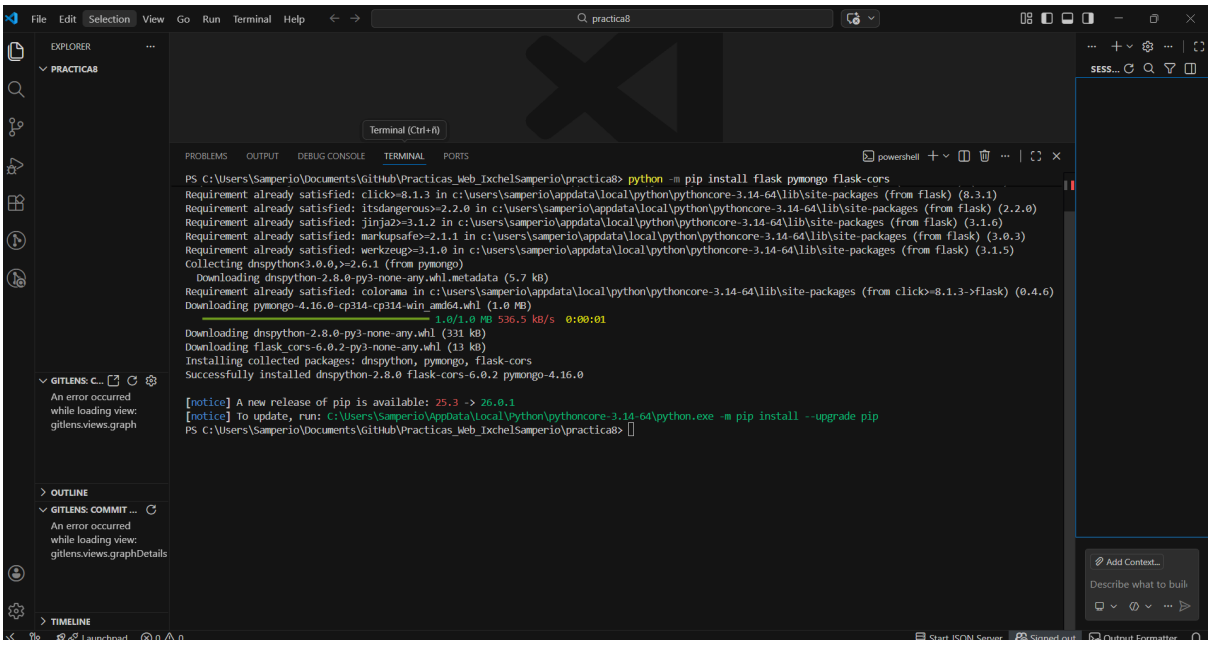
Universidad Politécnica de Pachuca
Enero-Abril 2026


MongoDB Compass - localhost:27017/cartelera_cultural.eventos
Connections Edit View Collection Help
Compass
My Queries
Data Modeling
CONNECTIONS (1)
Search connections
localhost:27017
admin
cartelera_cultural
eventos
config
local
Welcome
localhost:27017
eventos
localhost:27017 > cartelera_cultural > eventos
Open MongoDB shell
Documents 0
Aggregations
Schema
Indexes 1
Validation
Your pipeline is currently empty. Need help getting started?
Explain
Run
Options
eventos – modified
CREATE NEW
PREVIEW
0 Documents in the collection
Preview of documents
Add stage
Learn more about aggregation pipeline stages


MongoDB Compass - localhost:27017/cartelera_cultural.eventos
Connections Edit View Collection Help
Compass
My Queries
Data Modeling
CONNECTIONS (1)
Search connections
localhost:27017
admin
cartelera_cultural
eventos
config
local
Welcome
localhost:27017
eventos
localhost:27017 > cartelera_cultural > eventos
Open MongoDB shell
Documents 3
Aggregations
Schema
Indexes 1
Validation
{}
Explain
Reset
Find
Options
25
1 - 3 of 3
{
"_id": {...},
"titulo": "Concierto Sinfónico",
"fecha": "2026-03-15",
"lugar": "Teatro Hidalgo",
"precio": 250
}
{
"_id": {...},
"titulo": "Festival de Rock",
"fecha": "2026-04-20",
"lugar": "Auditorio Nacional",
"precio": 300
}
{
"_id": {...},
"titulo": "Obra de Teatro Clásica",
"fecha": "2026-05-10",
"lugar": "Casa de la Cultura",
"precio": 180
}


MongoDB Compass - localhost:27017/cartelera_cultural.eventos
Connections Edit View Collection Help
Compass
My Queries
Data Modeling
CONNECTIONS (1)
Search connections
localhost:27017
admin
cartelera_cultural
eventos
config
local
startup_log
Welcome
localhost:27017
cartelera_cultural
eventos
local
localhost:27017 > cartelera_cultural > eventos
Open MongoDB shell
Documents 3
Aggregations
Schema
Indexes 1
Validation
Generate query
Explain
Reset
Find
Options
ADD DATA
UPDATE
DELETE
EXPORT DATA
EXPORT CODE
25
1 - 3 of 3
{
"_id": {},
"titulo": "Concierto Sinfónico",
"fecha": "2026-03-15",
"lugar": "Teatro Hidalgo",
"precio": 250
}
{
"_id": {},
"titulo": "Festival de Rock",
"fecha": "2026-04-20",
"lugar": "Auditorio Nacional",
"precio": 300
}
{
"_id": {},
"titulo": "Obra de Teatro Clásica",
"fecha": "2026-05-10",
"lugar": "Casa de la Cultura",
"precio": 180
}


PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
powershell
PS C:\Users\Samperio\Documents\GitHub\Practicas_Web_IxchelSamperio\practica8>
PS C:\Users\Samperio\Documents\GitHub\Practicas_Web_IxchelSamperio\practica8> python -m pip install flask pymongo flask-cors
Python was not found; run without arguments to install from the Microsoft Store, or disable this shortcut from Settings > Apps > Advanced app settings > App execution aliases.
PS C:\Users\Samperio\Documents\GitHub\Practicas_Web_IxchelSamperio\practica8>


File Edit Selection View Go Run Terminal Help
practica8
EXPLORER
PRACTICA8
Terminal (Ctrl+ñ)
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
powershell
PS C:\Users\Samperio\Documents\GitHub\Practicas_Web_IxchelSamperio\practica8> python -m pip install flask pymongo flask-cors
Requirement already satisfied: click>=8.1.3 in c:\users\samperio\appdata\local\python\pythoncore-3.14-64\lib\site-packages (from flask) (8.3.1)
Requirement already satisfied: itsdangerous>=2.2.0 in c:\users\samperio\appdata\local\python\pythoncore-3.14-64\lib\site-packages (from flask) (2.2.0)
Requirement already satisfied: jinja2>=3.1.2 in c:\users\samperio\appdata\local\python\pythoncore-3.14-64\lib\site-packages (from flask) (3.1.6)
Requirement already satisfied: markupsafe>=2.1.1 in c:\users\samperio\appdata\local\python\pythoncore-3.14-64\lib\site-packages (from flask) (3.0.3)
Requirement already satisfied: werkzeug>=3.1.0 in c:\users\samperio\appdata\local\python\pythoncore-3.14-64\lib\site-packages (from flask) (3.1.5)
Collecting dnspython<3.0.0,>=2.6.1 (from pymongo)
Downloading dnspython-2.8.0-py3-none-any.whl.metadata (5.7 kB)
Requirement already satisfied: colorama in c:\users\samperio\appdata\local\python\pythoncore-3.14-64\lib\site-packages (from click>=8.1.3->flask) (0.4.6)
Downloading pymongo-4.16.0-cp314-cp314-win_amd64.whl (1.0 MB)
1.0/1.0 MB 536.5 kB/s 0:00:01
Downloading dnspython-2.8.0-py3-none-any.whl (331 kB)
Downloading flask_cors-6.0.2-py3-none-any.whl (13 kB)
Installing collected packages: dnspython, pymongo, flask-cors
Successfully installed dnspython-2.8.0 flask-cors-6.0.2 pymongo-4.16.0
[notice] A new release of pip is available: 25.3 -> 26.0.1
[notice] To update, run: C:\Users\Samperio\AppData\Local\Python\pythoncore-3.14-64\python.exe -m pip install --upgrade pip
PS C:\Users\Samperio\Documents\GitHub\Practicas_Web_IxchelSamperio\practica8>
GITLENS: C...
An error occurred while loading view: gitlens.views.graph
OUTLINE
GITLENS: COMMIT ...
An error occurred while loading view: gitlens.views.graphDetails
TIMELINE
Add Context...
Describe what to build

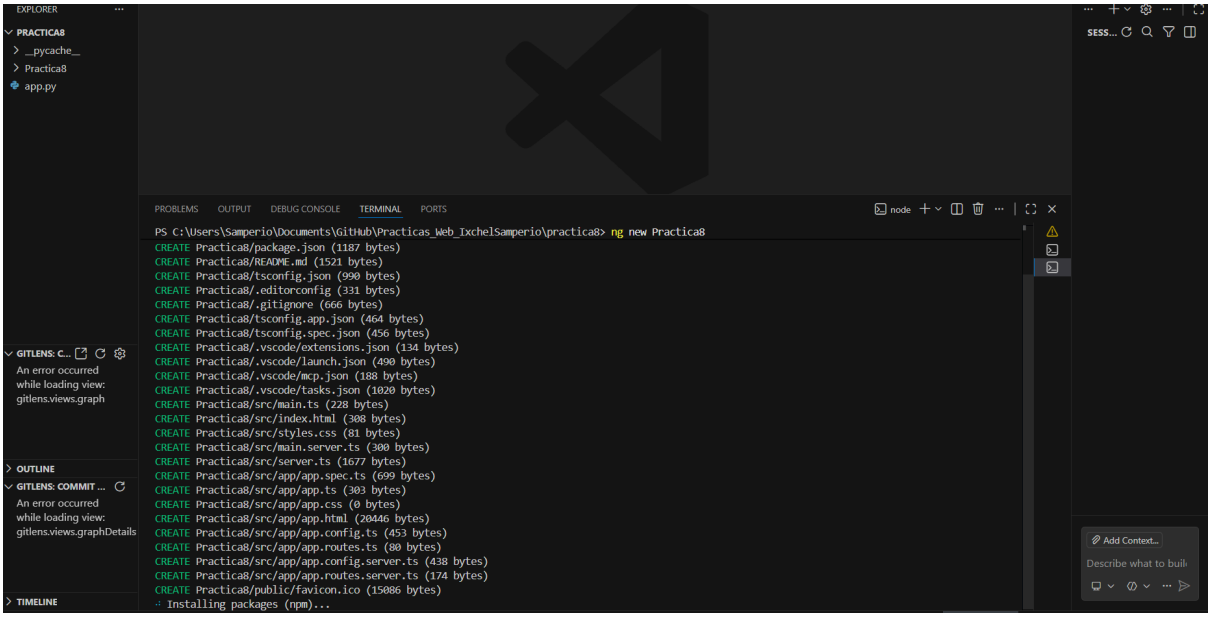
```python
    return "API Cartelera Cultural funcionando"

@app.route("/eventos", methods=["GET"])
def obtener_eventos():
    eventos = list(coleccion.find({}, {"_id": 0}))
    return jsonify(eventos)

if __name__ == "__main__":
    app.run(debug=True)
```

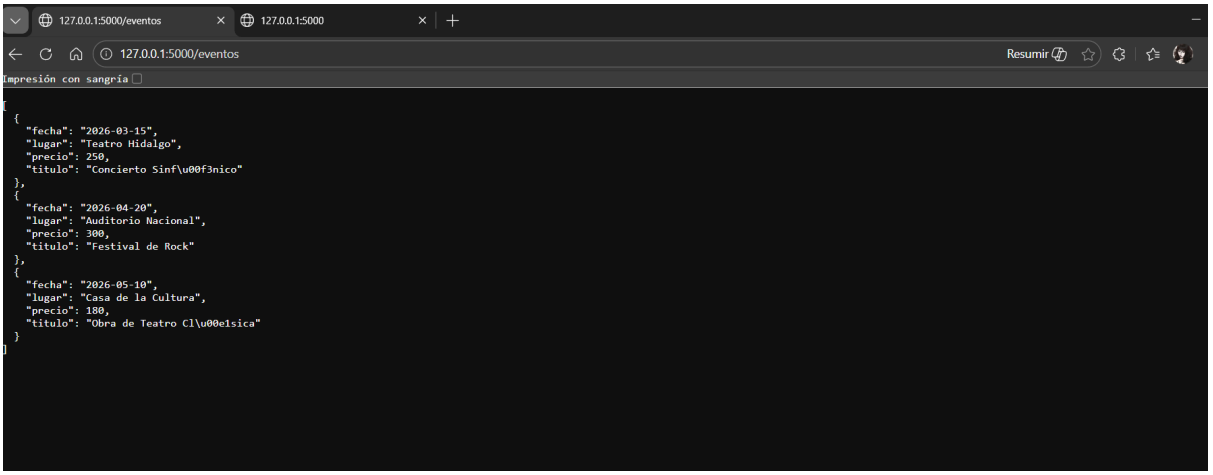
```
PS C:\Users\Samperio\Documents\GitHub\Practicas_Web_IxchelSamperio\practica8> cd python app.py
Set-Location : No se encuentra ningún parámetro de posición que acepte el argumento 'app.py'.
En línea: 1 Carácter: 1
+ cd python app.py
+ ~~~~~~~~~~~~~~~
    + CategoryInfo          : InvalidArgument: (:) [Set-Location], ParameterBindingException
    + FullyQualifiedErrorId : PositionalParameterNotFound,Microsoft.PowerShell.Commands.SetLocationCommand

PS C:\Users\Samperio\Documents\GitHub\Practicas_Web_IxchelSamperio\practica8> py app.py
 * Serving Flask app 'app'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 814-240-358
127.0.0.1 - - [19/Feb/2026 13:39:12] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [19/Feb/2026 13:39:12] "GET /favicon.ico HTTP/1.1" 404 -
```

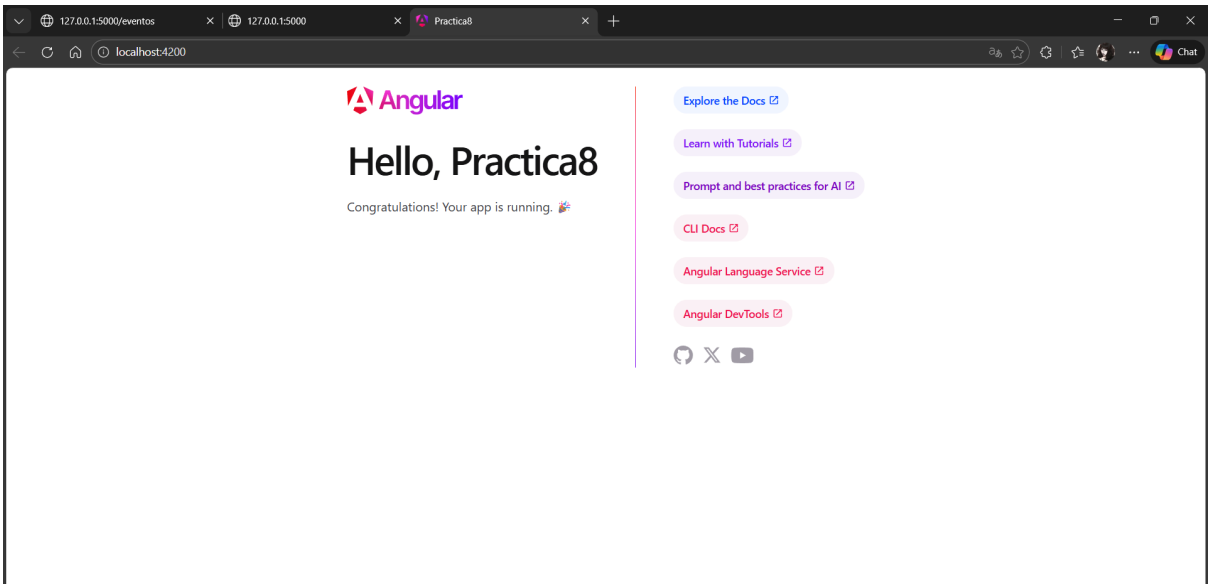
```python
from flask import Flask, jsonify
from flask_cors import CORS
from pymongo import MongoClient

app = Flask(__name__)
CORS(app)

# Conexión a MongoDB
cliente = MongoClient("mongodb://localhost:27017/")
db = cliente["cartelera_cultural"]
coleccion = db["eventos"]

@app.route("/")
def inicio():
    return "API Cartelera Cultural funcionando"

@app.route("/eventos", methods=["GET"])
def obtener_eventos():
    eventos = list(coleccion.find({}, {"_id": 0}))
    return jsonify(eventos)

if __name__ == "__main__":
    app.run(debug=True)
```

127.0.0.1:5000

API Cartelera Cultural funcionando


EXPLORER
PRACTICA8
__pycache__
Practica8
app.py
GITLENS: C...
An error occurred while loading view: gitlens.views.graph
OUTLINE
GITLENS: COMMIT ...
An error occurred while loading view: gitlens.views.graphDetails
TIMELINE
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
node
PS C:\Users\Samperio\Documents\GitHub\Practicas_Web_IxchelSamperio\practica8> ng new Practica8
CREATE Practica8/package.json (1187 bytes)
CREATE Practica8/README.md (1521 bytes)
CREATE Practica8/tsconfig.json (990 bytes)
CREATE Practica8/.editorconfig (331 bytes)
CREATE Practica8/.gitignore (666 bytes)
CREATE Practica8/tsconfig.app.json (464 bytes)
CREATE Practica8/tsconfig.spec.json (456 bytes)
CREATE Practica8/.vscode/extensions.json (134 bytes)
CREATE Practica8/.vscode/launch.json (490 bytes)
CREATE Practica8/.vscode/mcp.json (188 bytes)
CREATE Practica8/.vscode/tasks.json (1020 bytes)
CREATE Practica8/src/main.ts (228 bytes)
CREATE Practica8/src/index.html (308 bytes)
CREATE Practica8/src/styles.css (81 bytes)
CREATE Practica8/src/main.server.ts (300 bytes)
CREATE Practica8/src/server.ts (1677 bytes)
CREATE Practica8/src/app/app.spec.ts (699 bytes)
CREATE Practica8/src/app/app.ts (303 bytes)
CREATE Practica8/src/app/app.css (0 bytes)
CREATE Practica8/src/app/app.html (20446 bytes)
CREATE Practica8/src/app/app.config.ts (453 bytes)
CREATE Practica8/src/app/app.routes.ts (80 bytes)
CREATE Practica8/src/app/app.config.server.ts (438 bytes)
CREATE Practica8/src/app/app.routes.server.ts (174 bytes)
CREATE Practica8/public/favicon.ico (15086 bytes)
Installing packages (npm)...
Add Context...
Describe what to build


127.0.0.1:5000/eventos
127.0.0.1:5000
127.0.0.1:5000/eventos
Resumir
Impresión con sangría
[
{
"fecha": "2026-03-15",
"lugar": "Teatro Hidalgo",
"precio": 250,
"titulo": "Concierto Sinf\u00f3nico"
},
{
"fecha": "2026-04-20",
"lugar": "Auditorio Nacional",
"precio": 300,
"titulo": "Festival de Rock"
},
{
"fecha": "2026-05-10",
"lugar": "Casa de la Cultura",
"precio": 180,
"titulo": "Obra de Teatro Cl\u00e1sica"
}
]


127.0.0.1:5000/eventos
127.0.0.1:5000
Practica8
localhost:4200
Chat
Angular
Hello, Practica8
Congratulations! Your app is running. 🎉
Explore the Docs
Learn with Tutorials
Prompt and best practices for AI
CLI Docs
Angular Language Service
Angular DevTools

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                esbuild - Practica8

details and how to change this setting, see https://angular.dev/cli/analytics.

   No
Global setting: enabled
Local setting: disabled
Effective status: disabled
Browser bundles
Initial chunk files  | Names          |  Raw size
main.js              | main           |  48.49 kB |
styles.css           | styles         |  95 bytes |

                     | Initial total  |  48.59 kB


Server bundles
Initial chunk files  | Names          |  Raw size
main.server.mjs      | main.server    |  49.76 kB |
server.mjs           | server         |   1.81 kB |
polyfills.server.mjs | polyfills.server | 243 bytes |

Application bundle generation complete. [7.469 seconds] - 2026-02-19T19:47:20.025Z

Watch mode enabled. Watching for file changes...
NOTE: Raw file sizes do not reflect development server per-request transformations.
  → Local:   http://localhost:4200/
  → press h + enter to show help
0                                                                  Start JSON Server   Signed out
```

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                powershell - Practica8
PS C:\Users\Samperio\Documents\GitHub\Practicas_Web_IxchelSamperio\practica8> ng generate component tabla --standalone
Error: This command is not available when running the Angular CLI outside a workspace.
PS C:\Users\Samperio\Documents\GitHub\Practicas_Web_IxchelSamperio\practica8> ng generate component tabla --standalone
Error: This command is not available when running the Angular CLI outside a workspace.
PS C:\Users\Samperio\Documents\GitHub\Practicas_Web_IxchelSamperio\practica8> cd Practica8
PS C:\Users\Samperio\Documents\GitHub\Practicas_Web_IxchelSamperio\practica8\Practica8> ng generate component tabla --standalone
CREATE src/app/tabla/tabla.spec.ts (547 bytes)
CREATE src/app/tabla/tabla.ts (193 bytes)
CREATE src/app/tabla/tabla.css (0 bytes)
CREATE src/app/tabla/tabla.html (21 bytes)
PS C:\Users\Samperio\Documents\GitHub\Practicas_Web_IxchelSamperio\practica8\Practica8>
```

```
File  Edit  Selection  View  Go  Run  Terminal  Help                                  practica8
EXPLORER                                  tabla.ts   tabla.html   app.ts
PRACTICA8                                 Practica8 > src > app > tabla > tabla.html > ...
  __pycache__                             Go to component
  Practica8                               1  <div class="container mt-4">
    .angular                              2    <h2>Cartelera Cultural</h2>
    .vscode                               3
    node_modules                          4    <table class="table table-bordered table-striped mt-3">
    public                                5      <thead class="table-dark">
    src                                   6        <tr>
      app                                 7          <th>Titulo</th>
        tabla                             8          <th>Fecha</th>
          tabla.css                       9          <th>Lugar</th>
          tabla.html                      10         <th>Categoria</th>
          tabla.spec.ts                   11         <th>Precio</th>
          tabla.ts                        12       </tr>
      app.config.server.ts                13     </thead>
      app.config.ts                       14
      app.css                             15     <tbody>
      app.html                            16       <tr *ngFor="let evento of listaEventos">
      app.routes.server.ts                17         <td>{{ evento.titulo }}</td>
      app.routes.ts                       18         <td>{{ evento.fecha }}</td>
      app.spec.ts                         19         <td>{{ evento.lugar }}</td>
      app.ts                              20         <td>{{ evento.categoria }}</td>
      index.html                          21         <td>
      main.server.ts                      22           <span *ngIf="evento.precio > 0">
      main.ts                             23             ${{ evento.precio }}
      server.ts                           24           </span>
      styles.css                          25           <span *ngIf="evento.precio == 0">
    .editorconfig                         26             Gratis
    .gitignore                            27           </span>
  GITLENS: COMMIT GRAPH                   28         </td>
  OUTLINE                                 29       </tr>
  GITLENS: COMMIT GRAPH...                30     </tbody>
  TIMELINE                                31   </table>
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                powershell
PS C:\Users\Samperio\Documents\GitHub\Practicas_Web_IxchelSamperio\practica8>
```

```
import { Component, OnInit } from '@angular/core';
import { HttpClient } from '@angular/common/http';
import { CommonModule } from '@angular/common';

@Component({
  selector: 'app-tabla',
  standalone: true,
  imports: [CommonModule],
  templateUrl: './tabla.html'
})
export class Tabla implements OnInit {

  listaEventos: any[] = [];

  constructor(private http: HttpClient) {}

  ngOnInit() {
    this.http.get<any[]>('http://127.0.0.1:5000/eventos')
      .subscribe(data => {
        this.listaEventos = data;
      });
  }
}
```

Cartelera Cultural

Título	Fecha	Lugar	Categoría	Precio
Concierto Sinfónico	2026-03-15	Teatro Hidalgo		$250
Festival de Rock	2026-04-20	Auditorio Nacional		$300
Obra de Teatro Clásica	2026-05-10	Casa de la Cultura		$180

Cartelera Cultural

Título	Fecha	Lugar	Categoría	Precio
Concierto Sinfónico	2026-03-15	Teatro Hidalgo		$250
Festival de Rock	2026-04-20	Auditorio Nacional		$300
Obra de Teatro Clásica	2026-05-10	Casa de la Cultura		$180
CIIMF	2026-02-20	UPP		